

Laboratory for Ecological Inference || Social Cognitive Neuroscience Lab

Introducing Lesion Profile Mapping

Jax Skye, PhD

October 21, 2025

INC Working Group Meeting

Outline

Background

Approach development

Lesion profile mapping pipeline

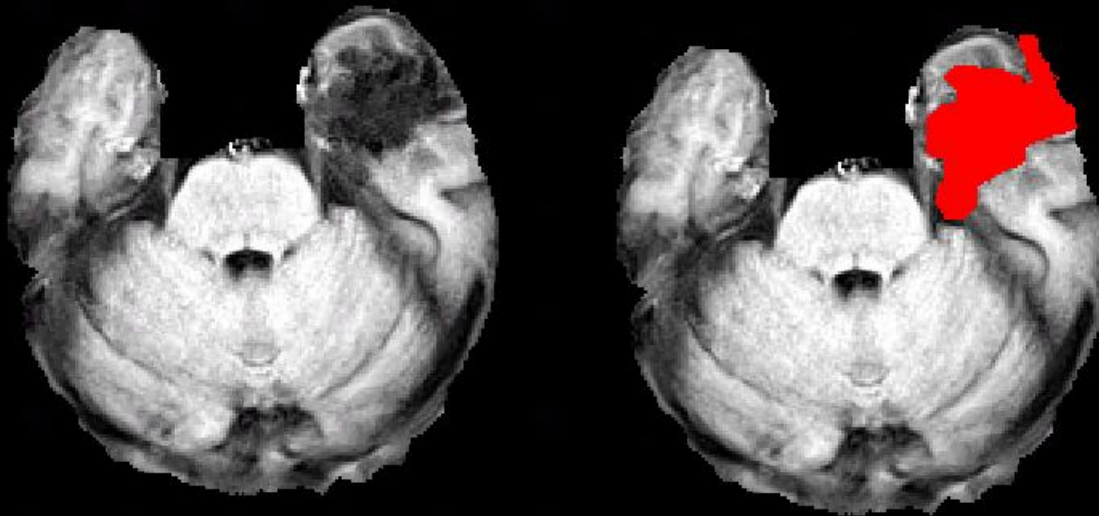
Future directions

Background

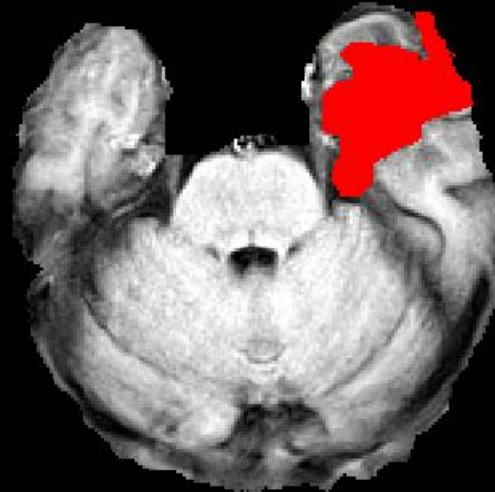
Lesion-Symptom Mapping



Lesion-Symptom Mapping

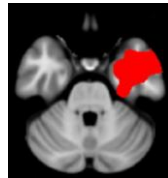


Lesion-Symptom Mapping

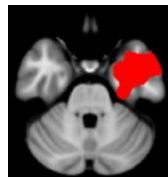


Brain lesion location (structural MRI)

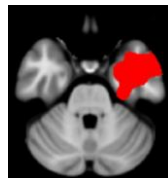
Behavioral test score



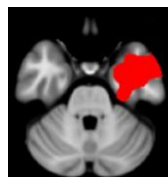
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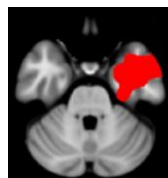
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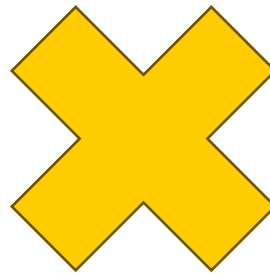
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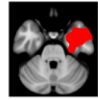


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Brain lesion location
(structural MRI)

Behavioral test
score



38

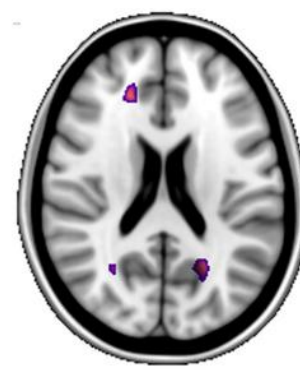
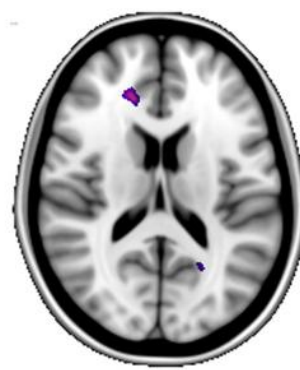
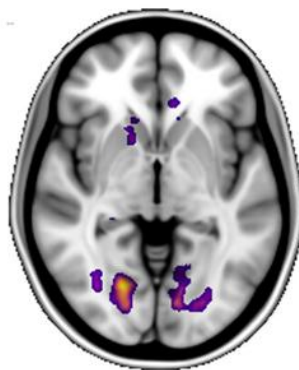
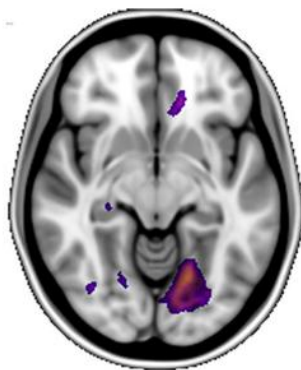
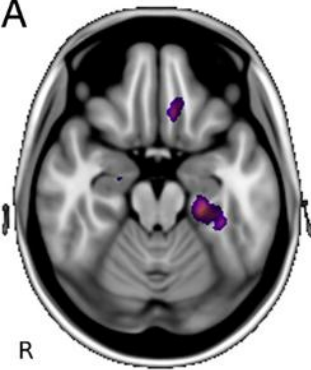
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22

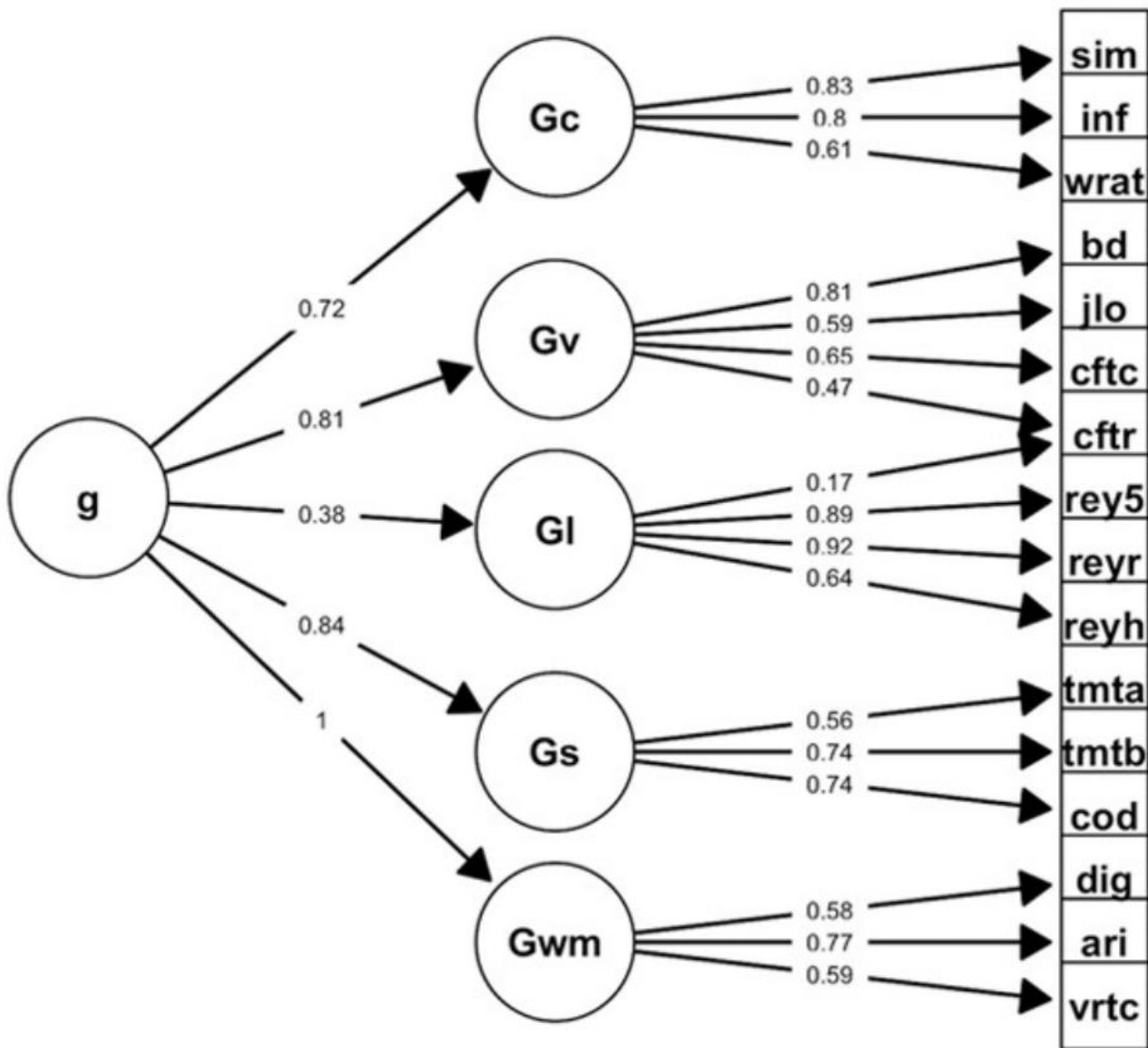
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A



(Skye et al, 2023)

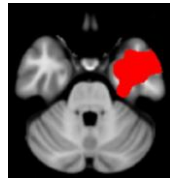
IOWA



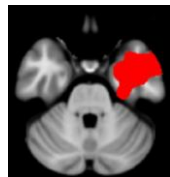
(Bowren et al, 2020)

Brain lesion location (structural MRI)

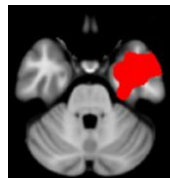
~~Behavioral test
score~~



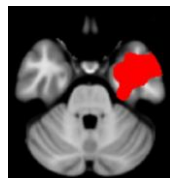
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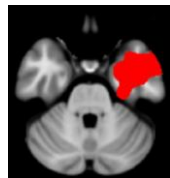
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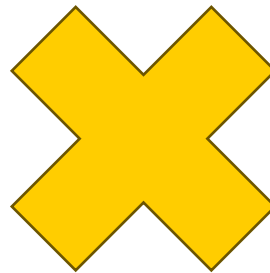
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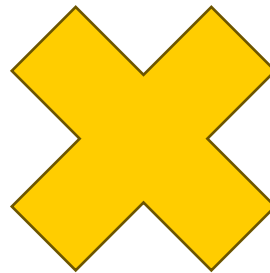
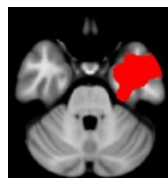
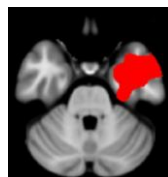
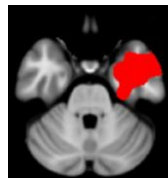
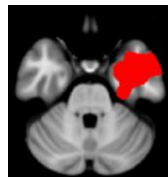
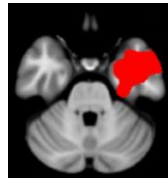


34



Brain lesion location (structural MRI)

Factor Score



1.1

-2.4

1.4

-0.2

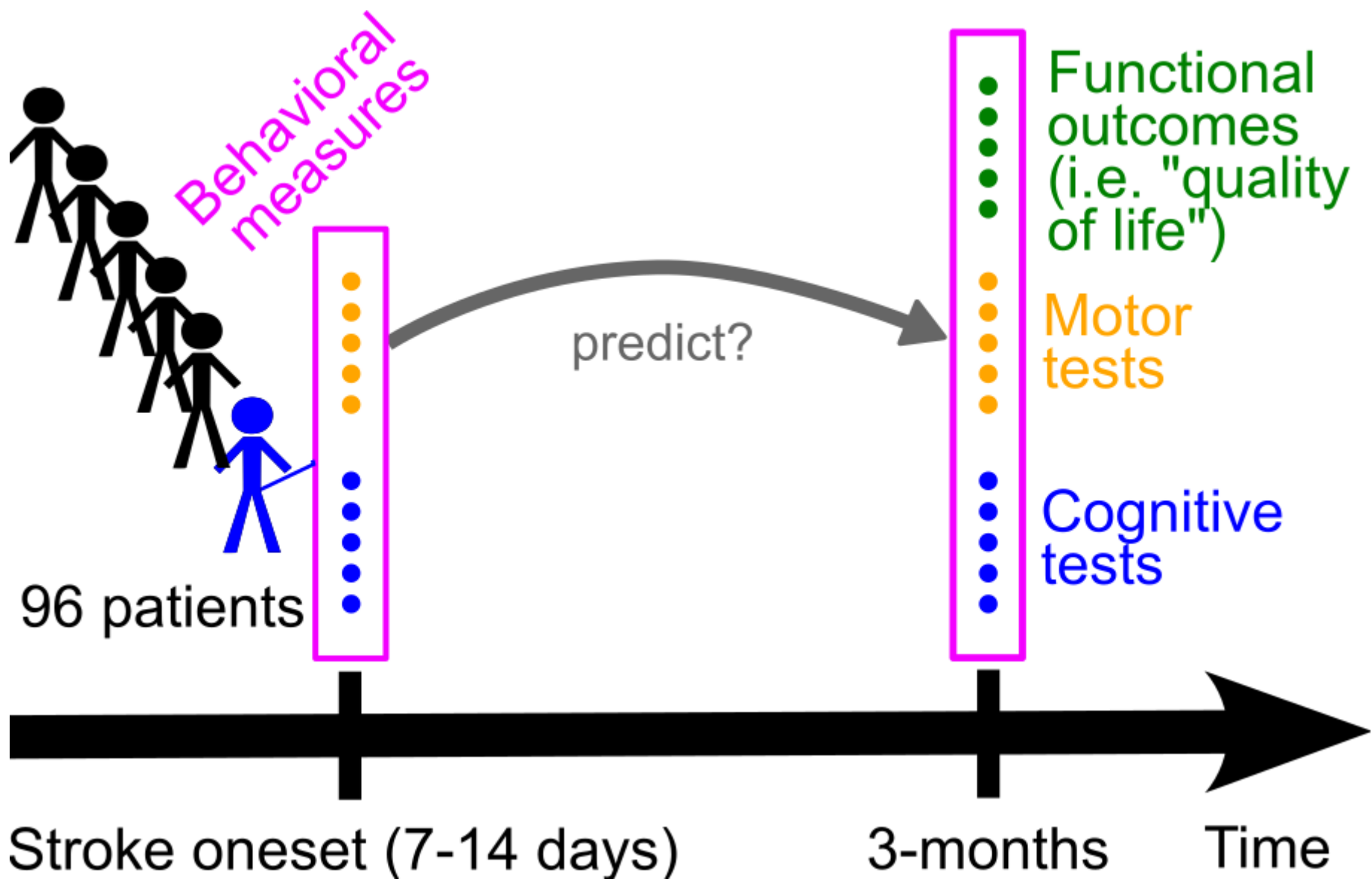
2.1



Methodological Gap

Mapping behavioral profiles

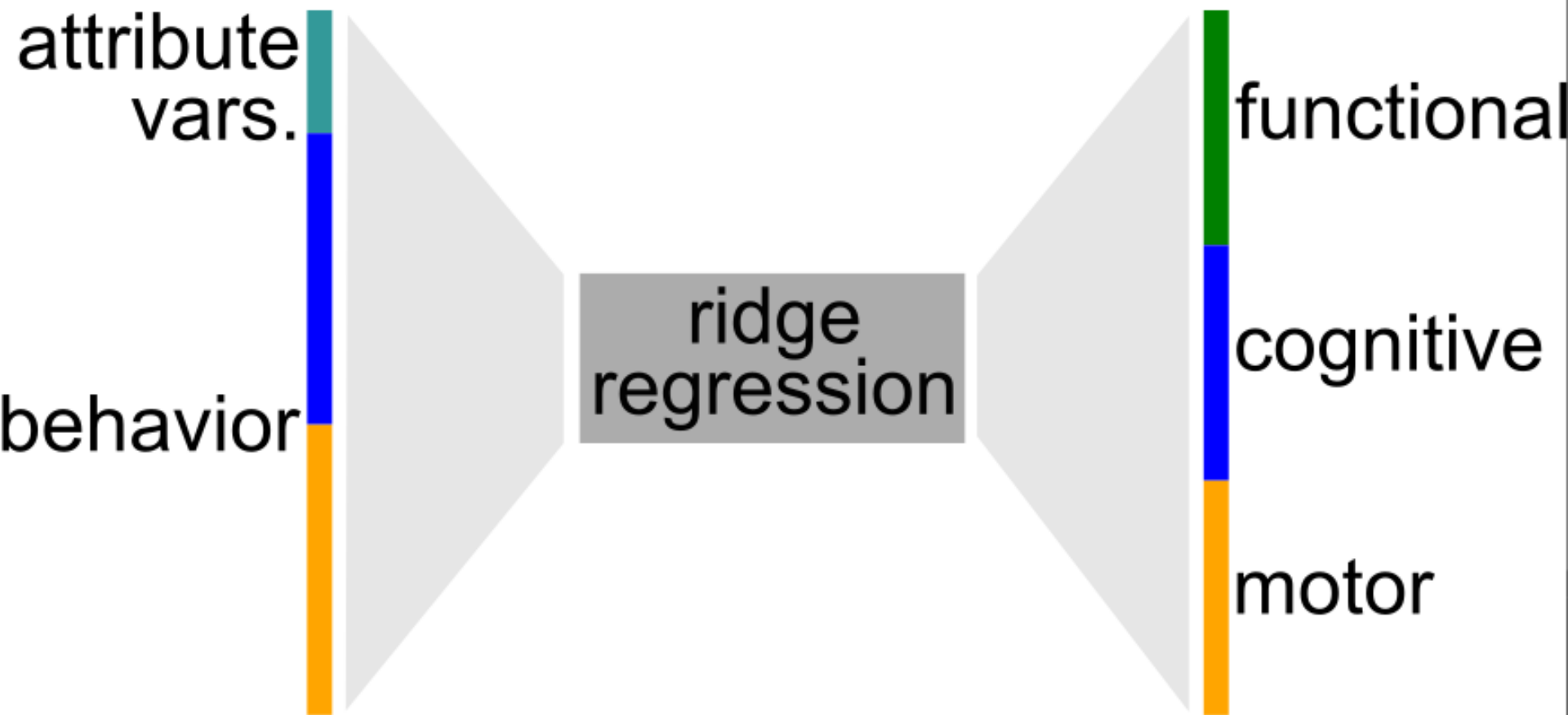
Approach Development



(Skye et al, *in prep*)

acute
predictors

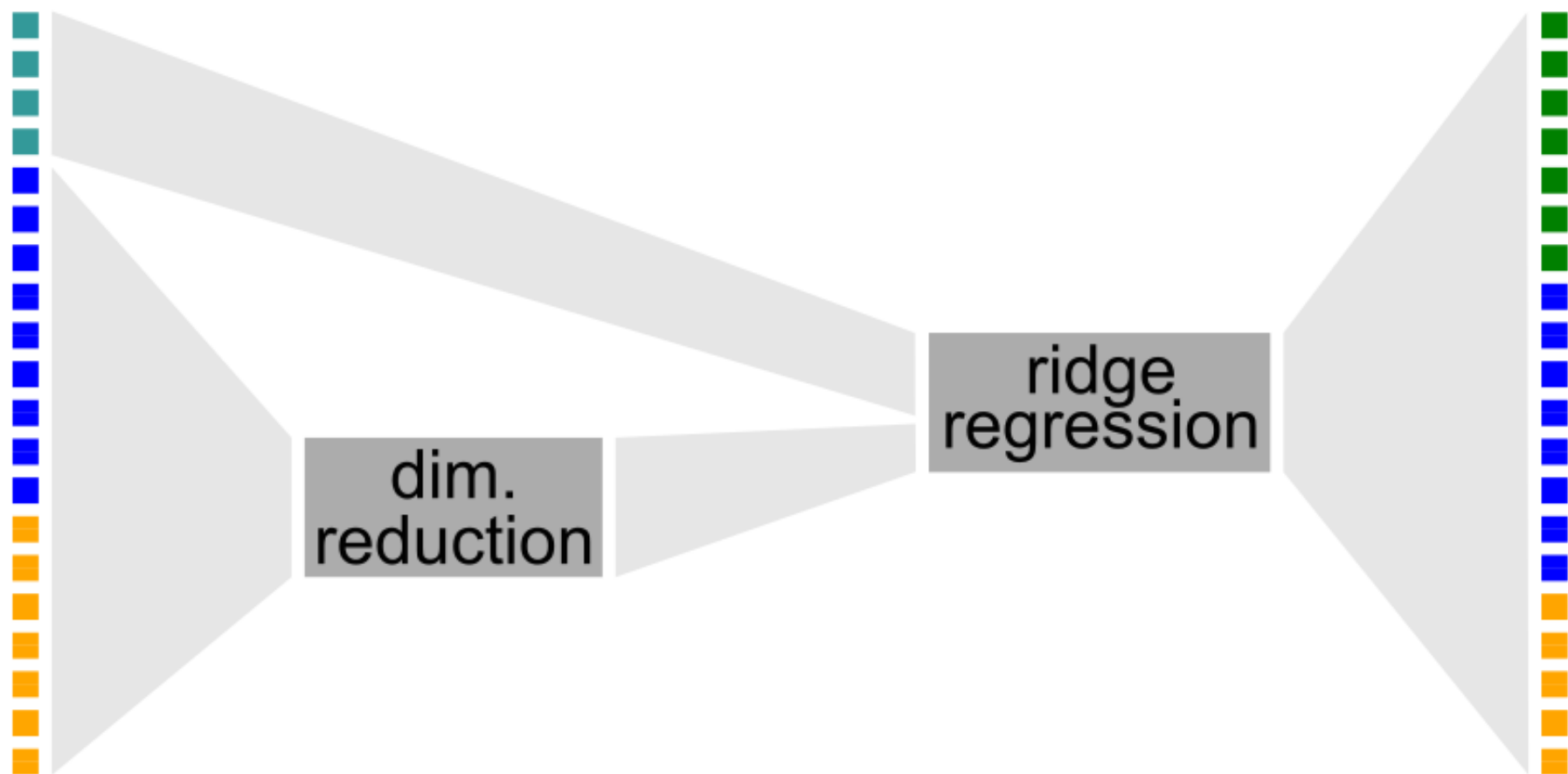
chronic
outcomes



(Skye et al, *in prep*)

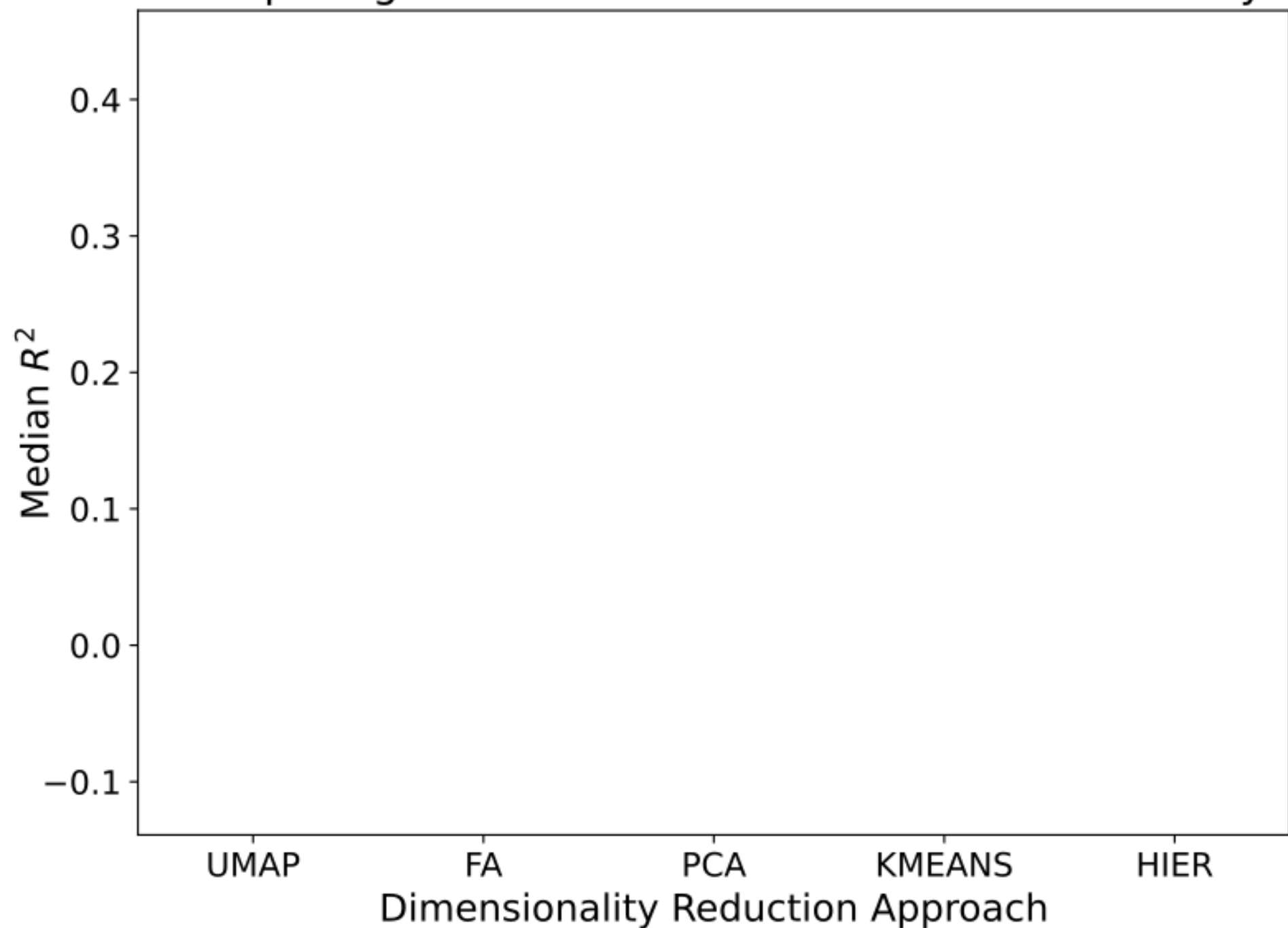
acute
predictors

chronic
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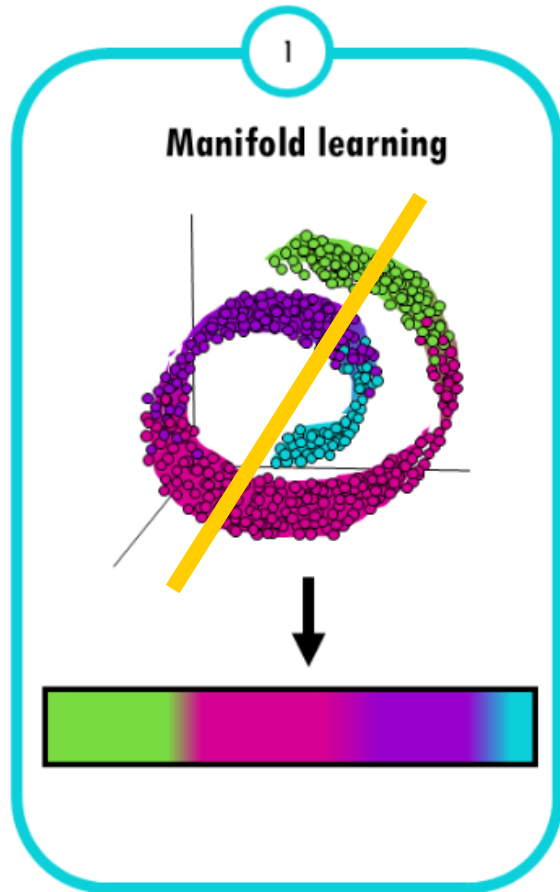


(Skye et al, *in prep*)

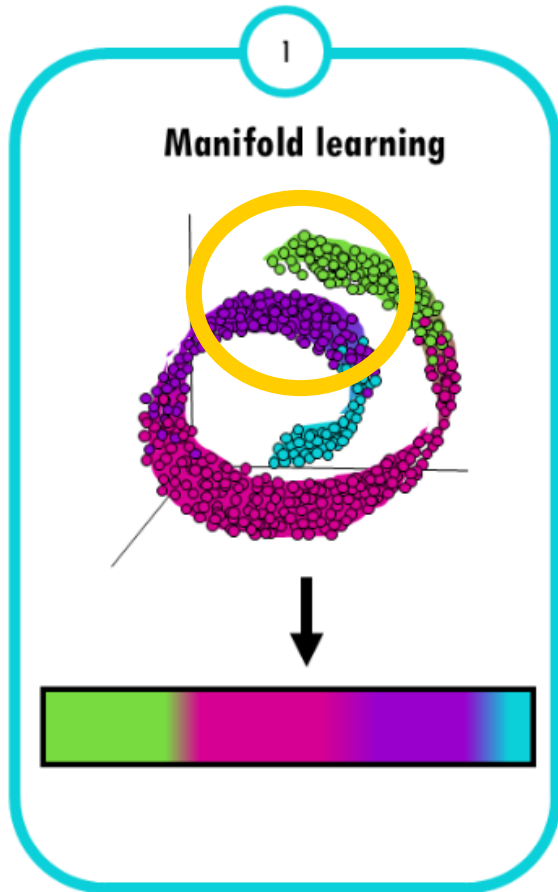
Comparing Functional Outcome Prediction Accuracy



How does UMAP summarise many dimensions into 2?



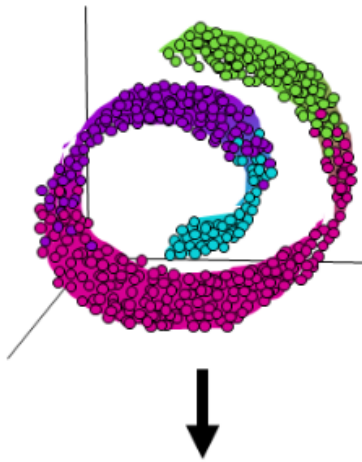
How does UMAP summarise many dimensions into 2?



How does UMAP summarise many dimensions into 2?

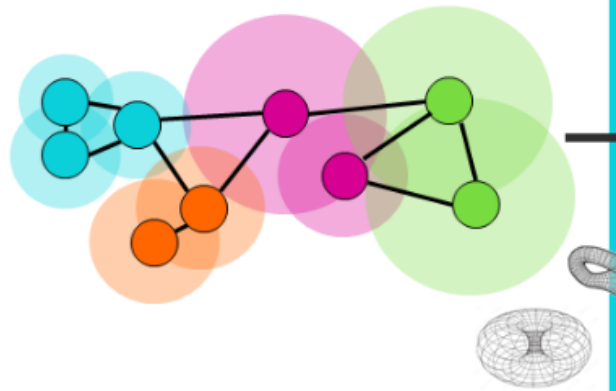
1

Manifold learning



2

Construct a high-dimensional graph

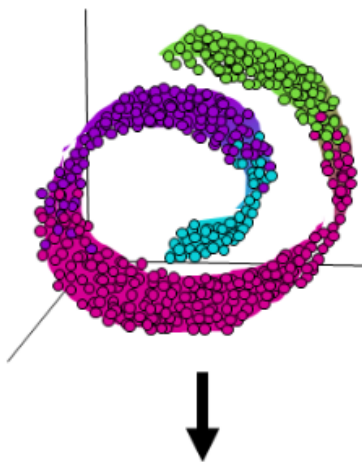


Fuzzy graph (multi-D)

How does UMAP summarise many dimensions into 2?

1

Manifold learning



2

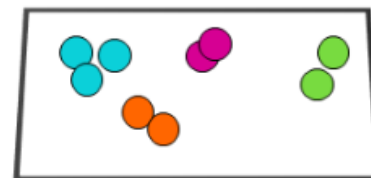
Construct a high-dimensional graph



Fuzzy graph (multi-D)

3

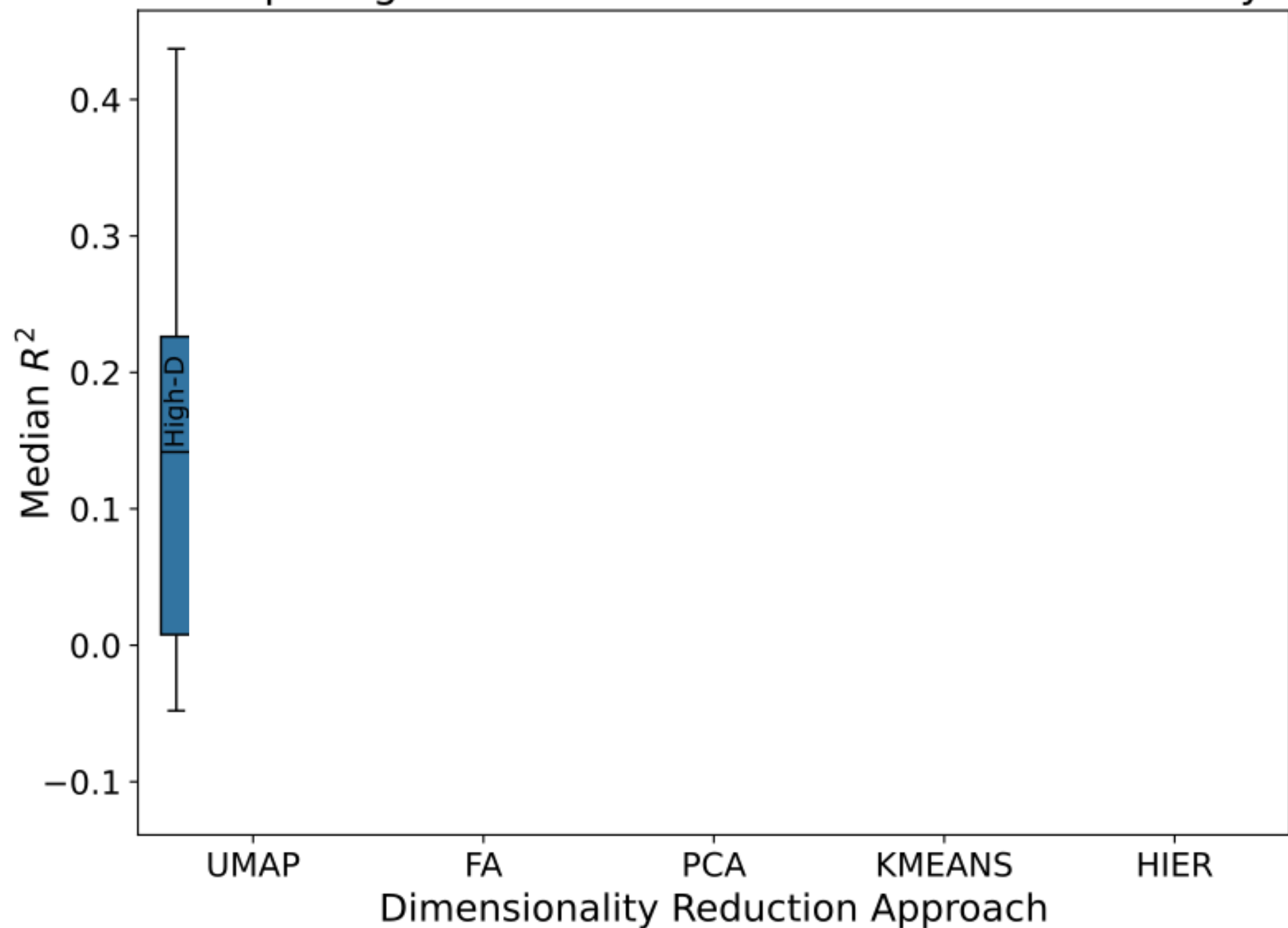
Project & optimise the graph in low dimensions



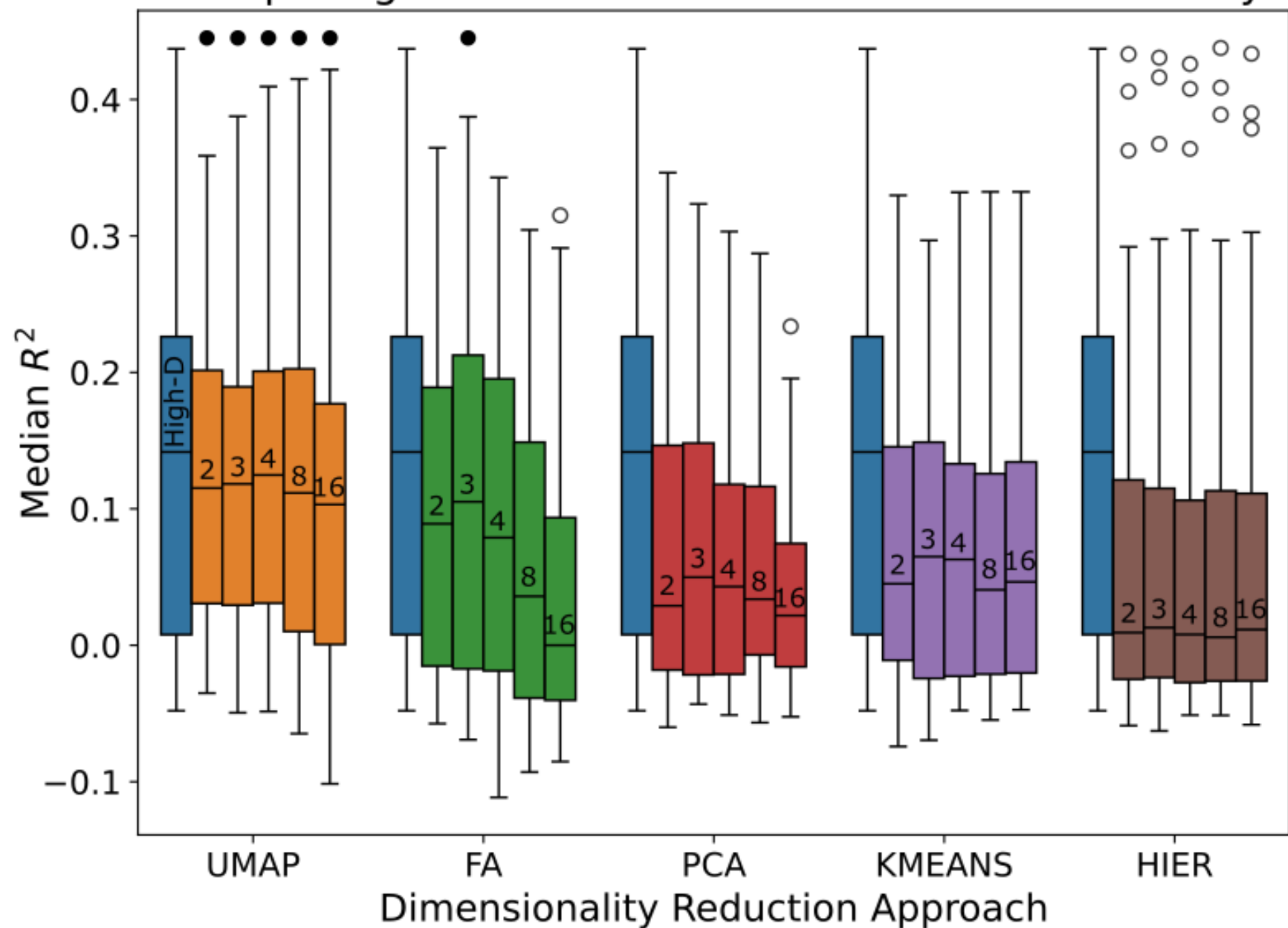
Keep local & global relationships

2D

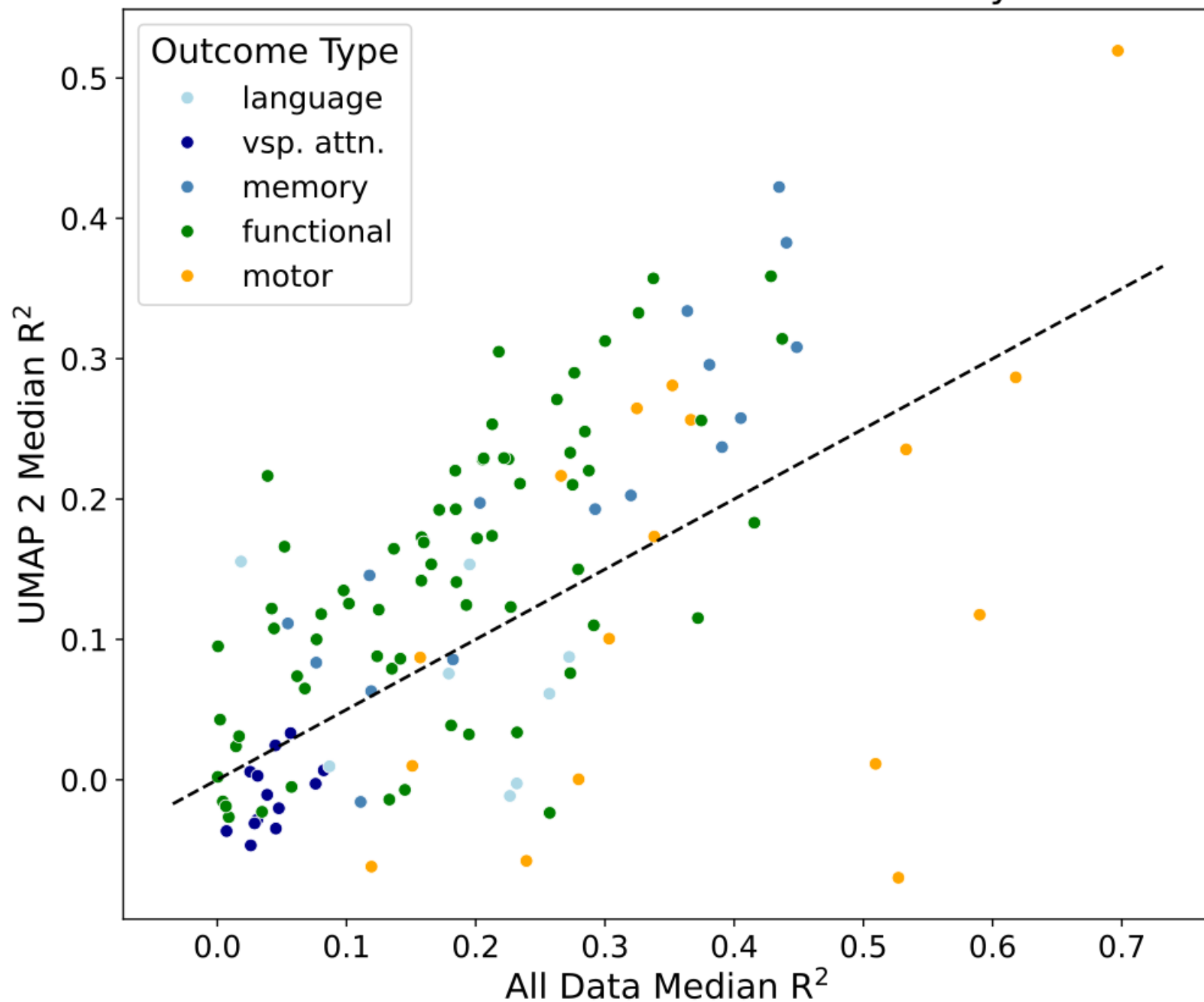
Comparing Functional Outcome Prediction Accuracy



Comparing Functional Outcome Prediction Accuracy



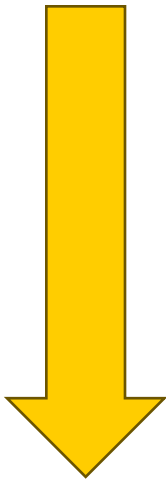
All Data v UMAP 2 Prediction Accuracy



Lesion Profile Mapping Pipeline

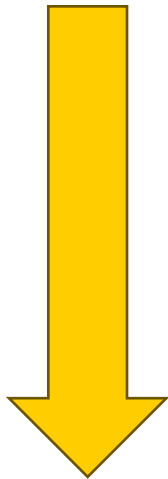
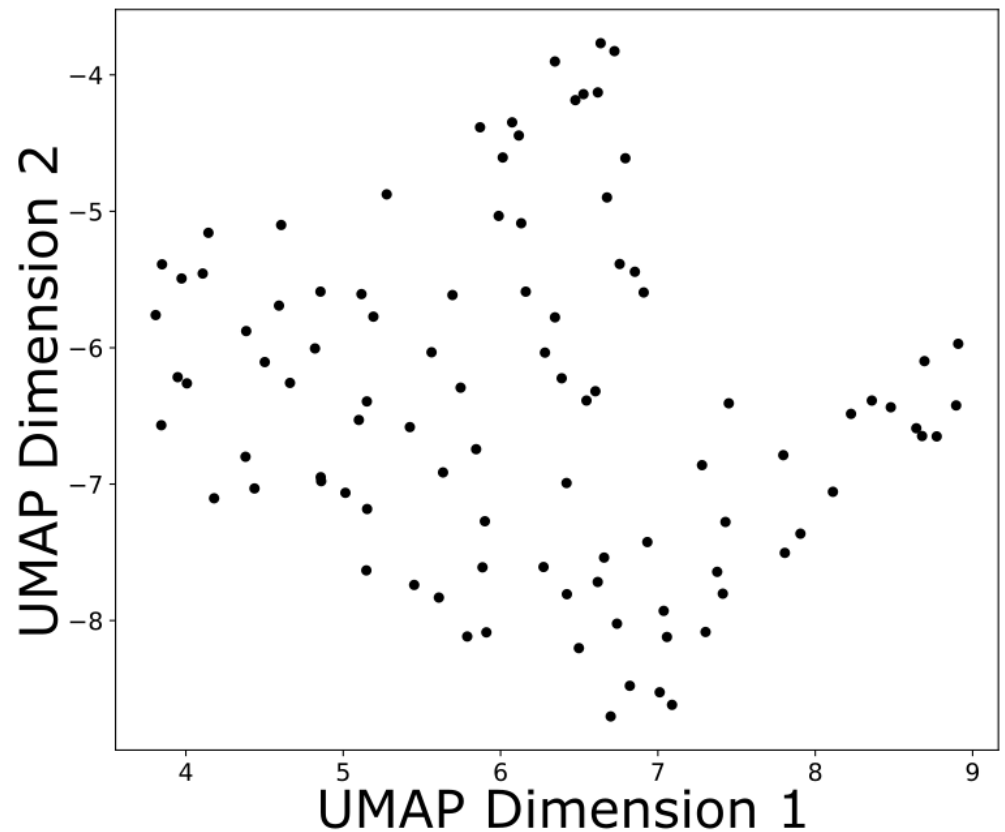
Example: WashU Dataset, subacute timepoint

Patient ID	BNT (T_1)	HVLT (T_2)	BVMT (T_3)	...	T_N
001					
002					
003					
...					
P_N					



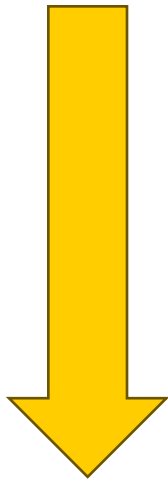
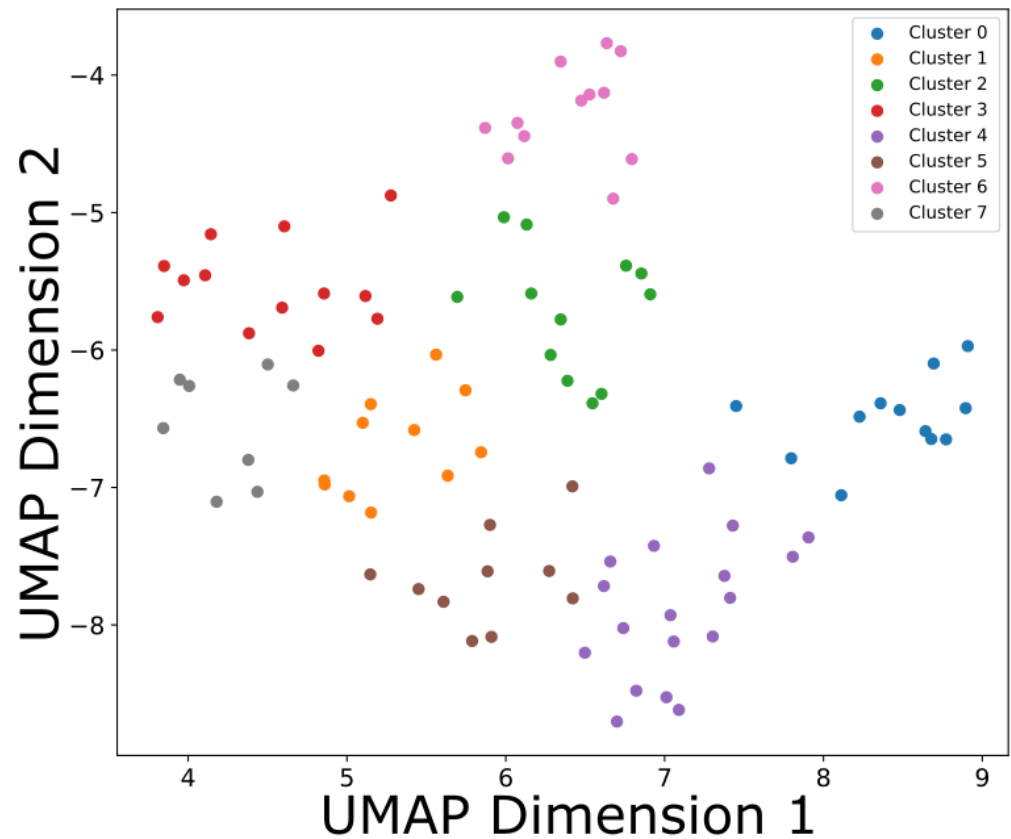
UMAP dimensionality
reduction on behavior

Patients plotted in
2D UMAP behavior
space



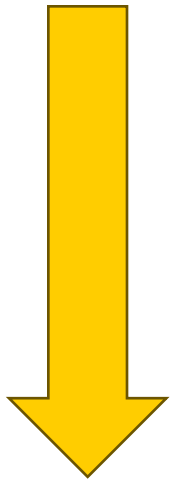
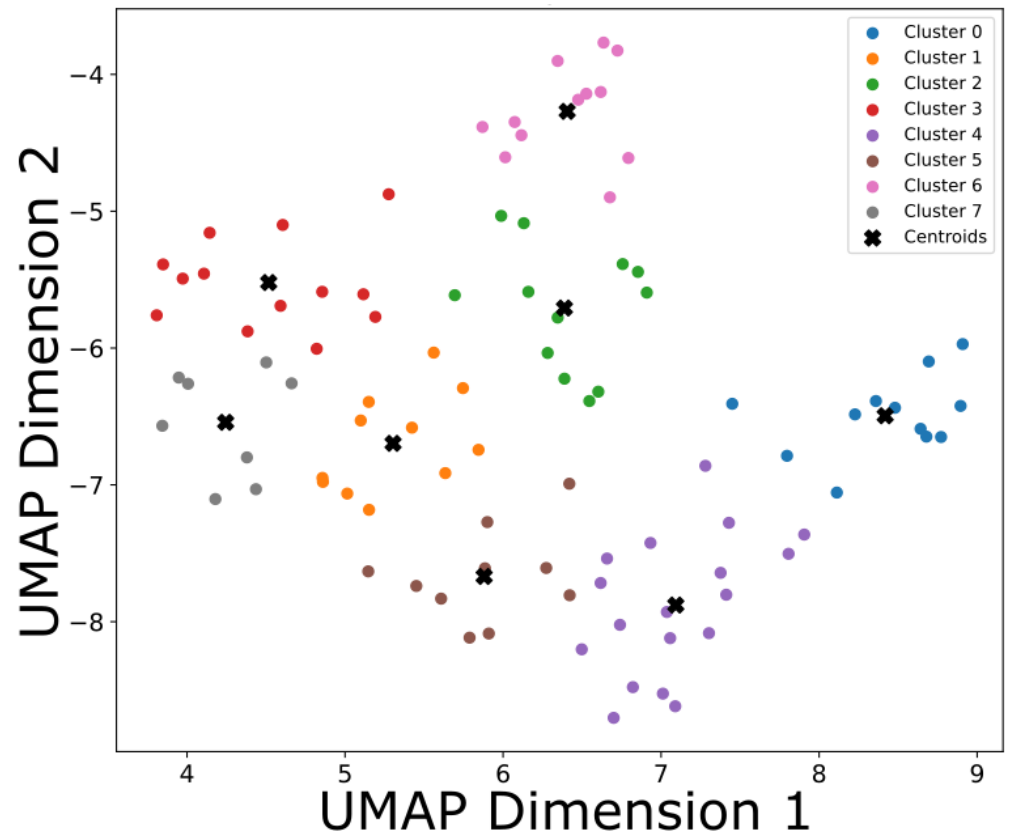
K-Means clustering on
UMAP embedding

Clusters determined
quantitatively or
qualitatively

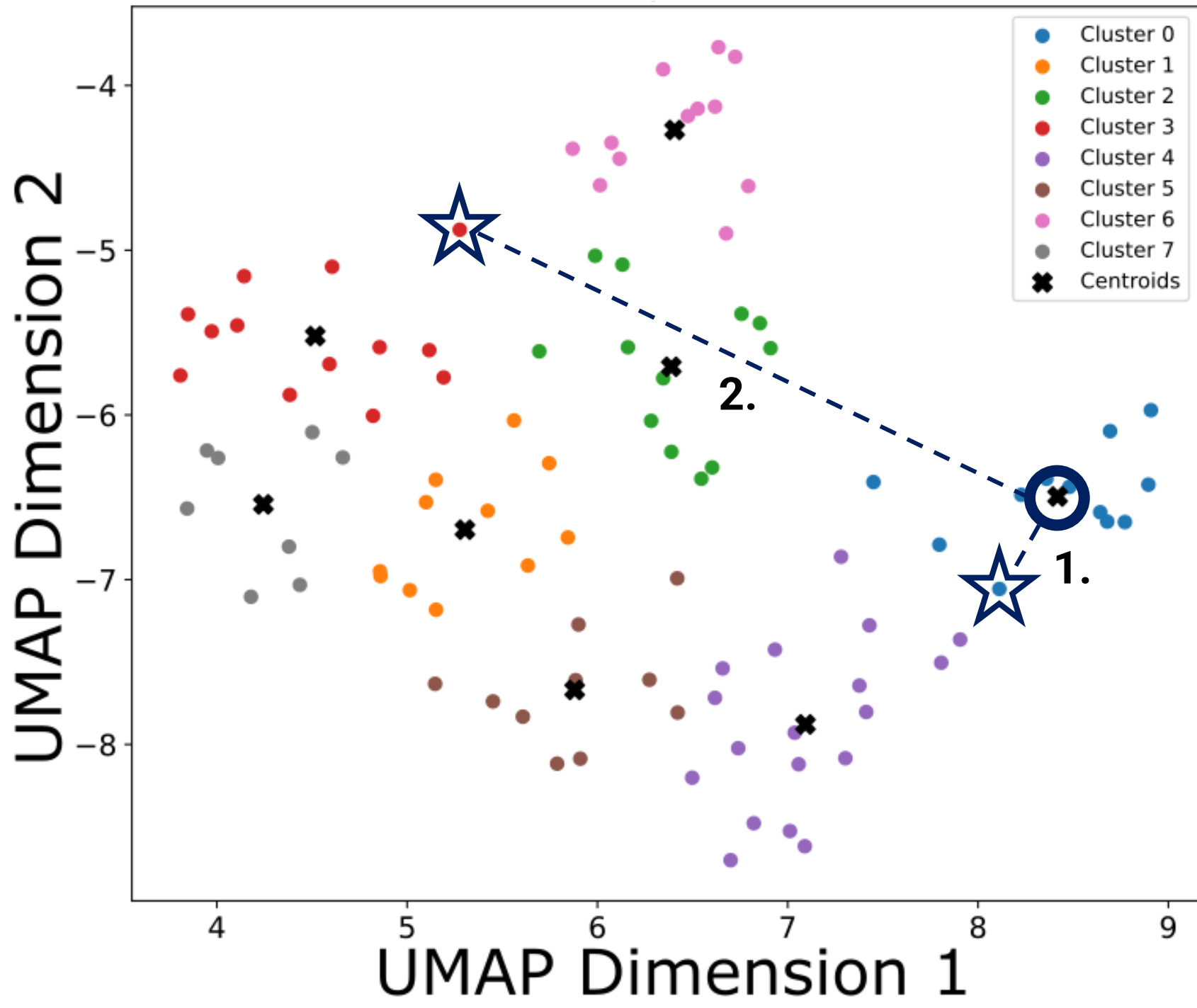


Calculate cluster
centroids

Centroids represent
location of a
behavioral profile in
behavior space



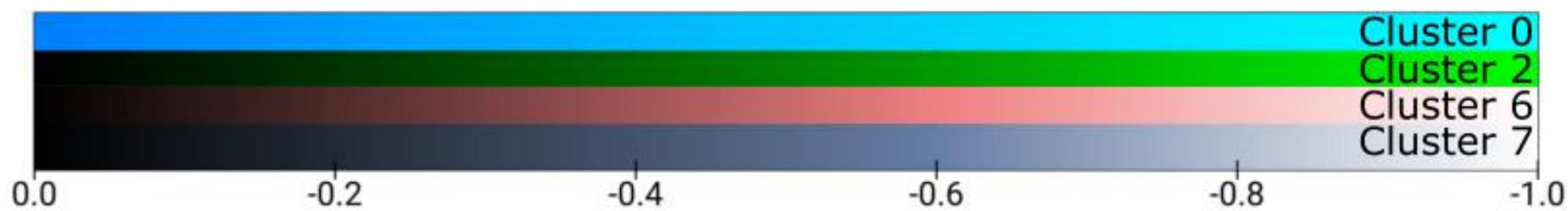
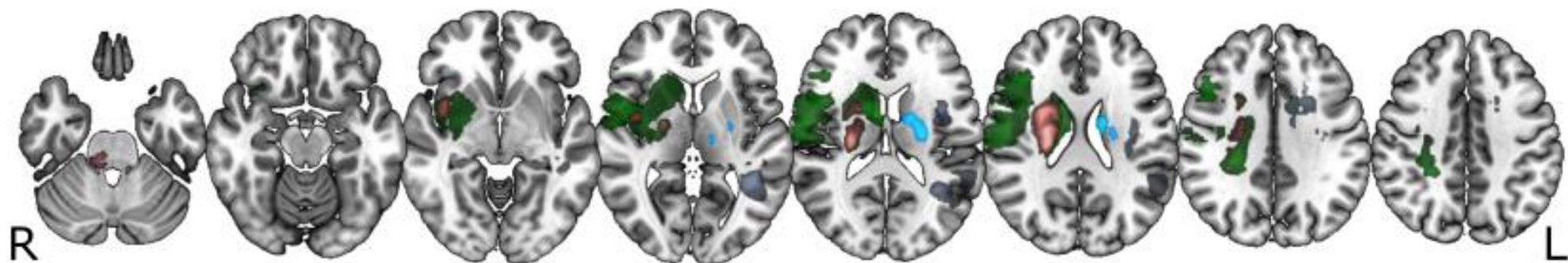
Calculate Euclidean
distance from each
patient to cluster centroid

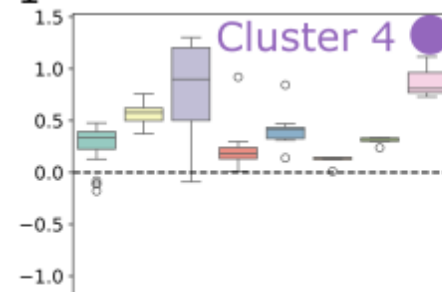
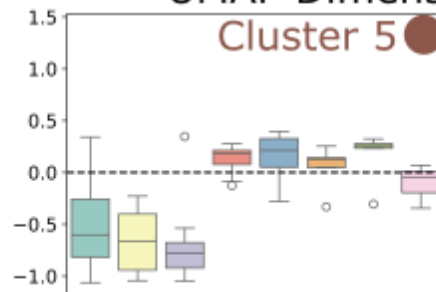
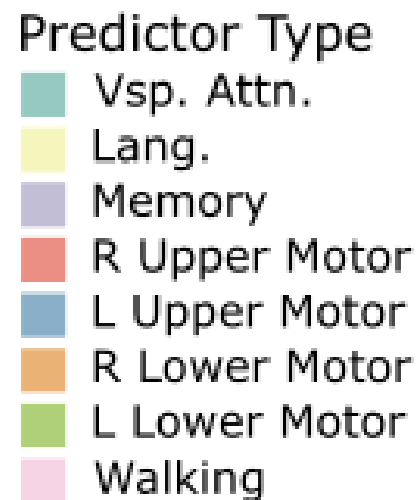
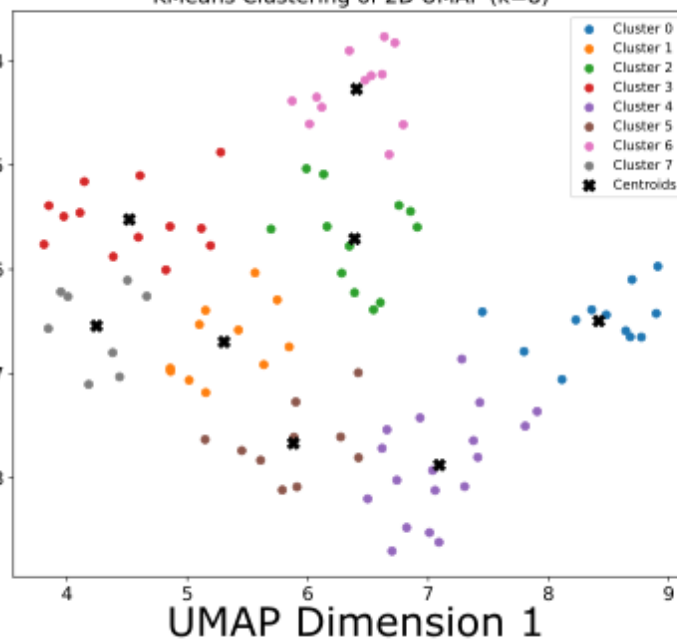
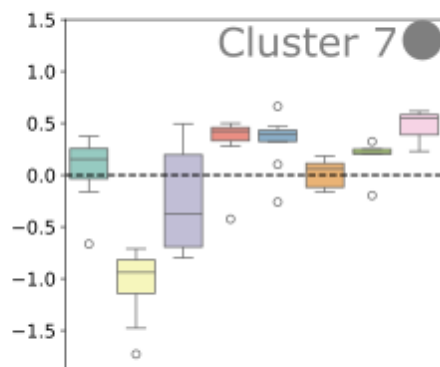
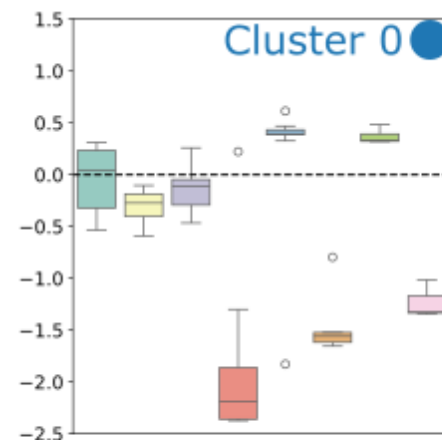
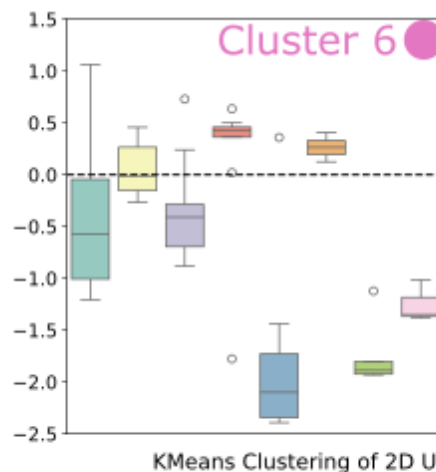
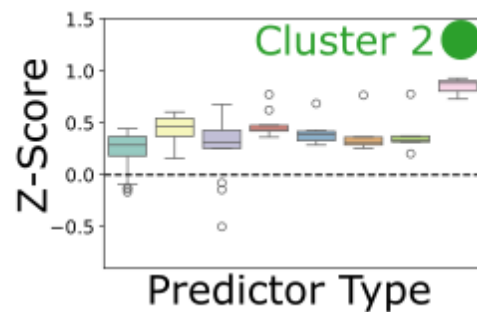


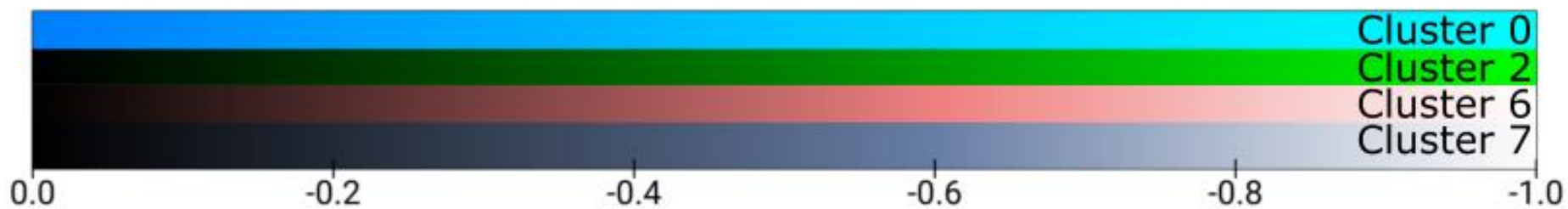
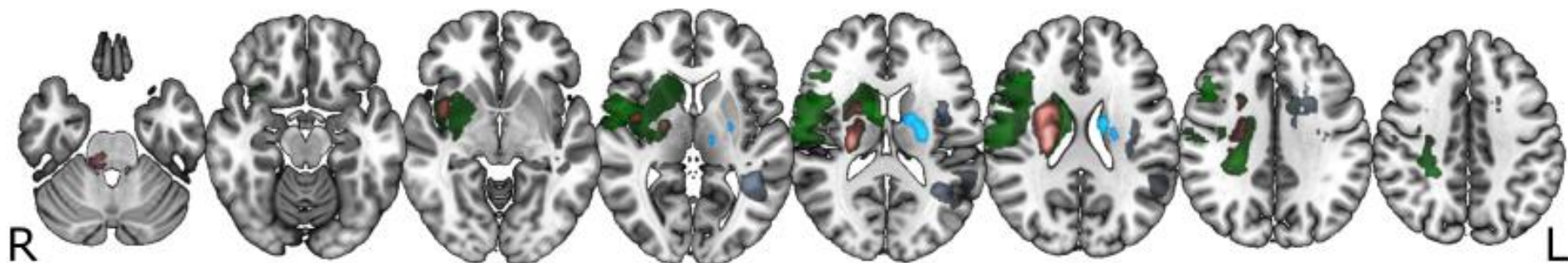
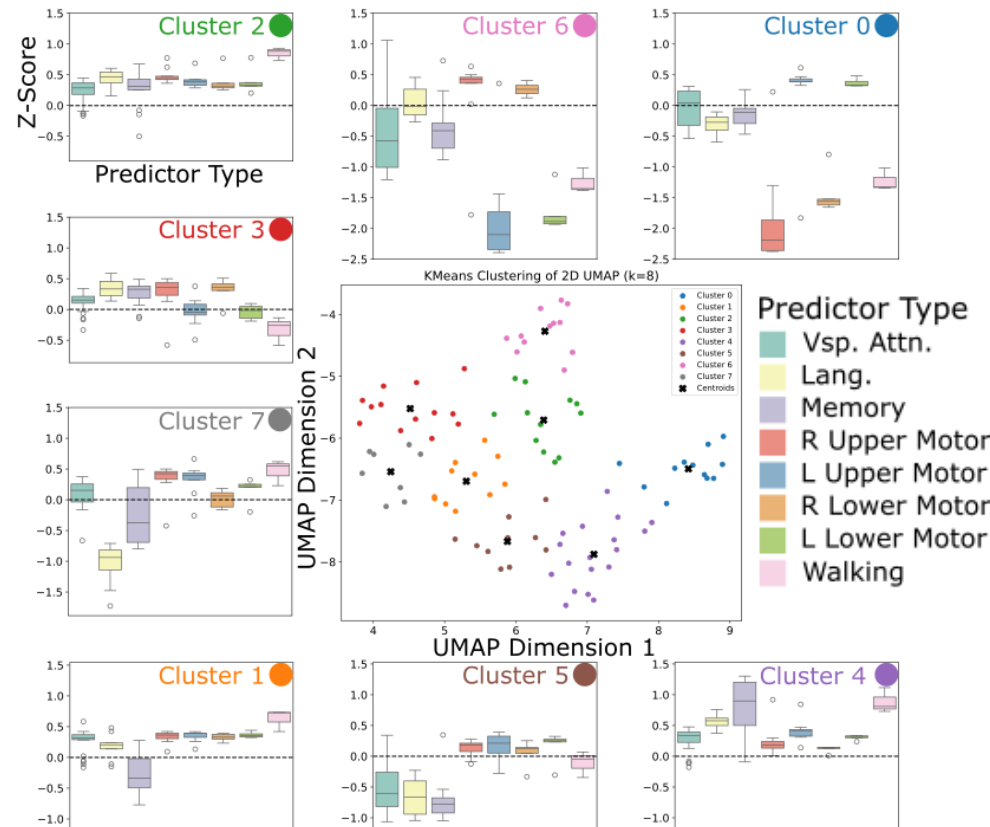
Output is New Lesion-Symptom Mapping Input

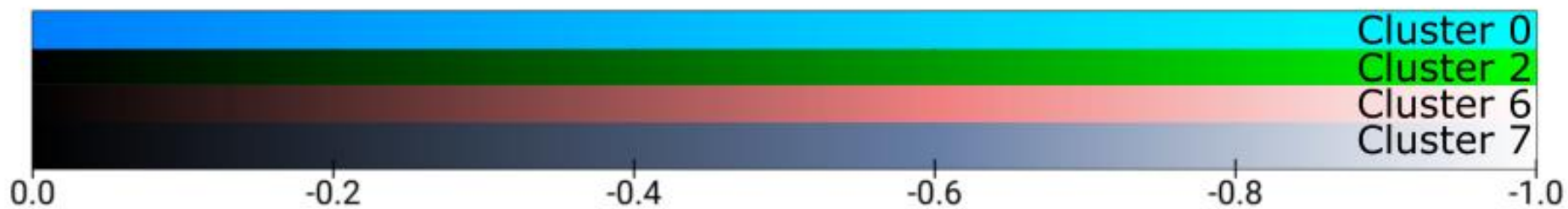
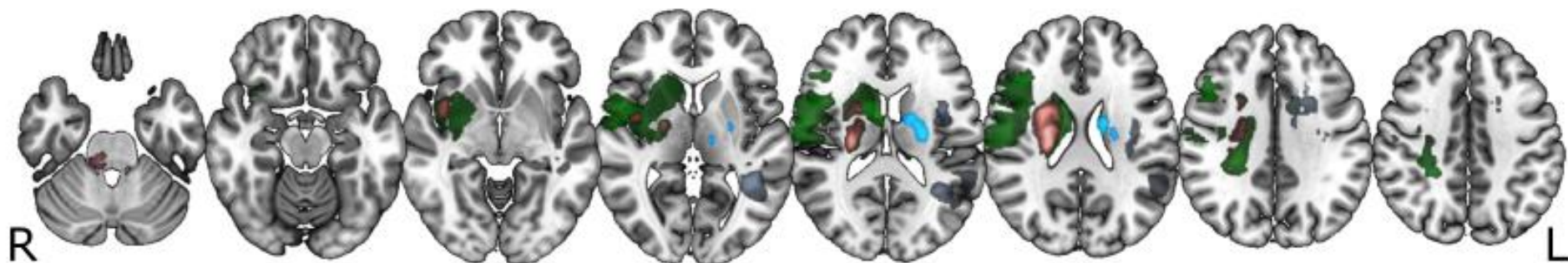
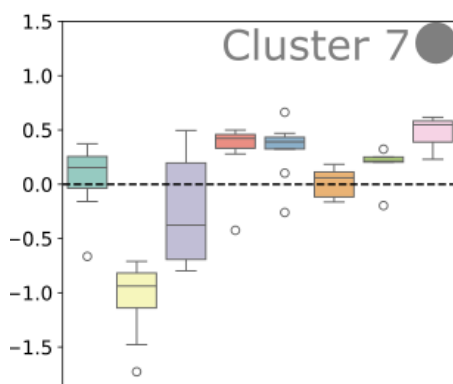
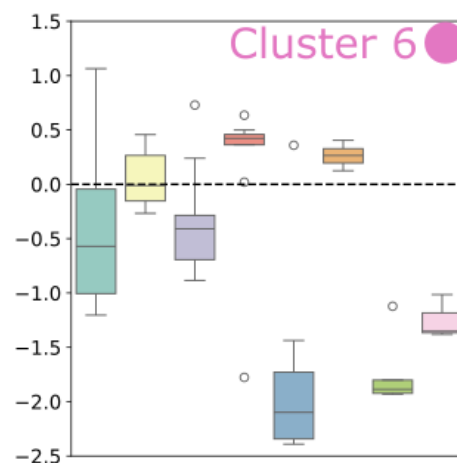
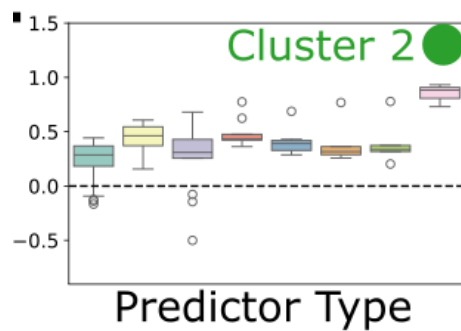
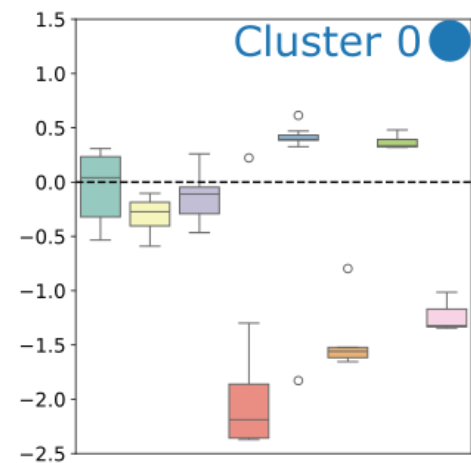
Patient ID	Distance to Cluster 0	Distance to Cluster 1	Distance to Cluster 2	...	Distance to Cluster N
001					
002					
003					
...					
P_N					

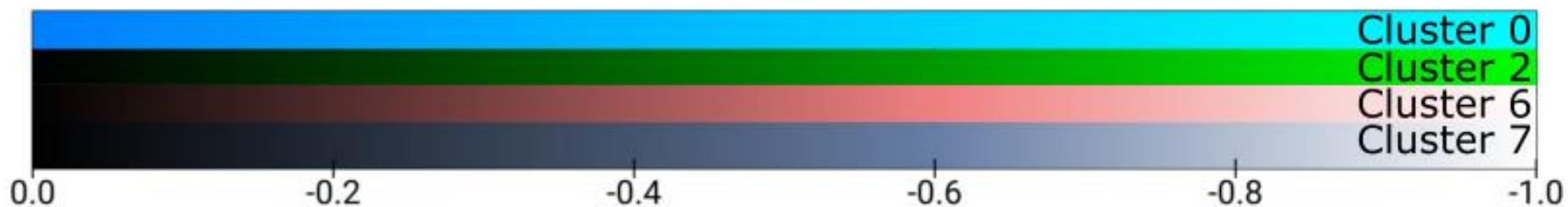
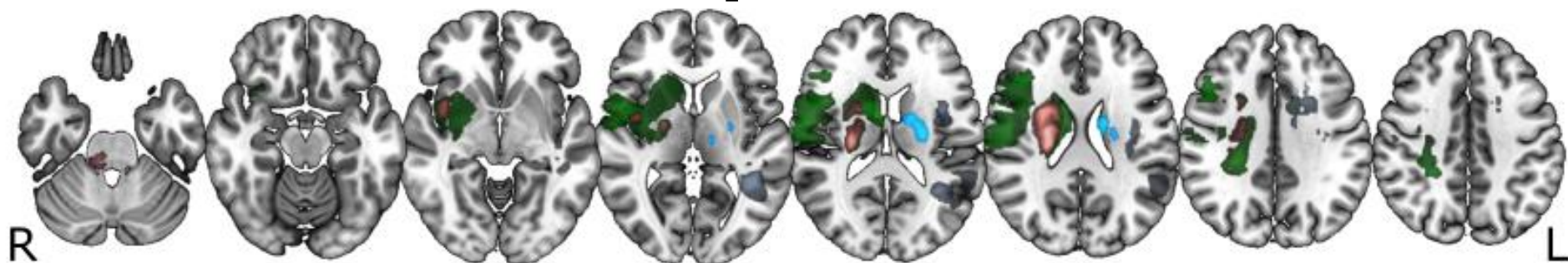
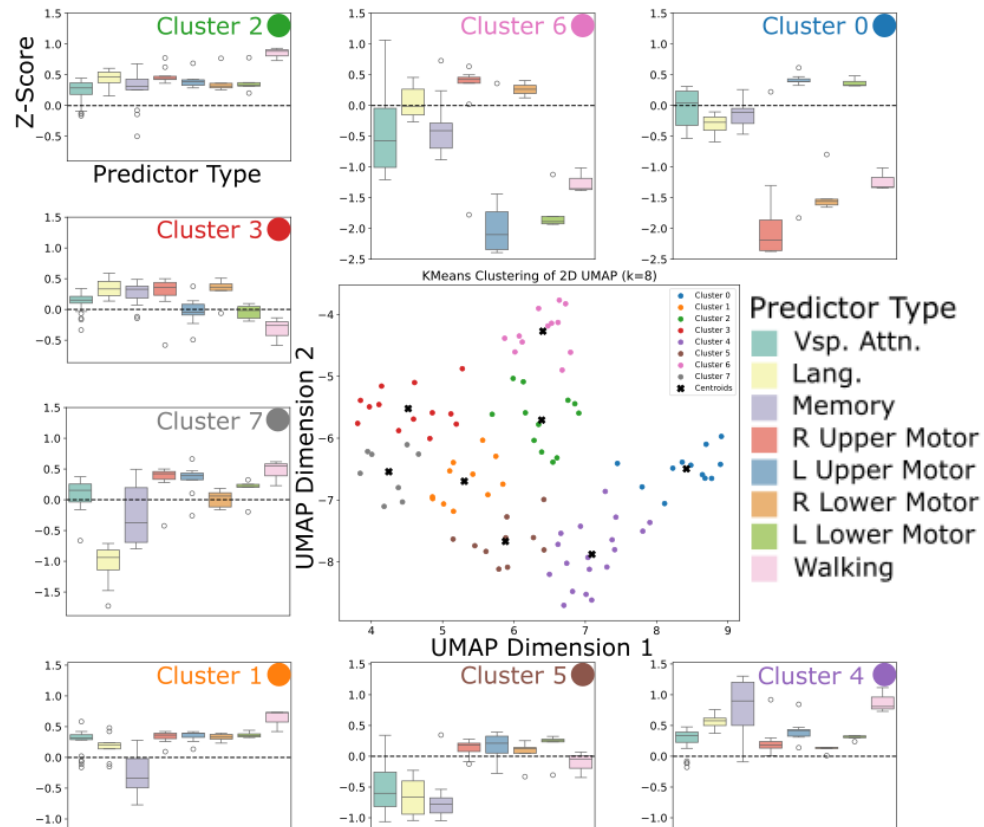
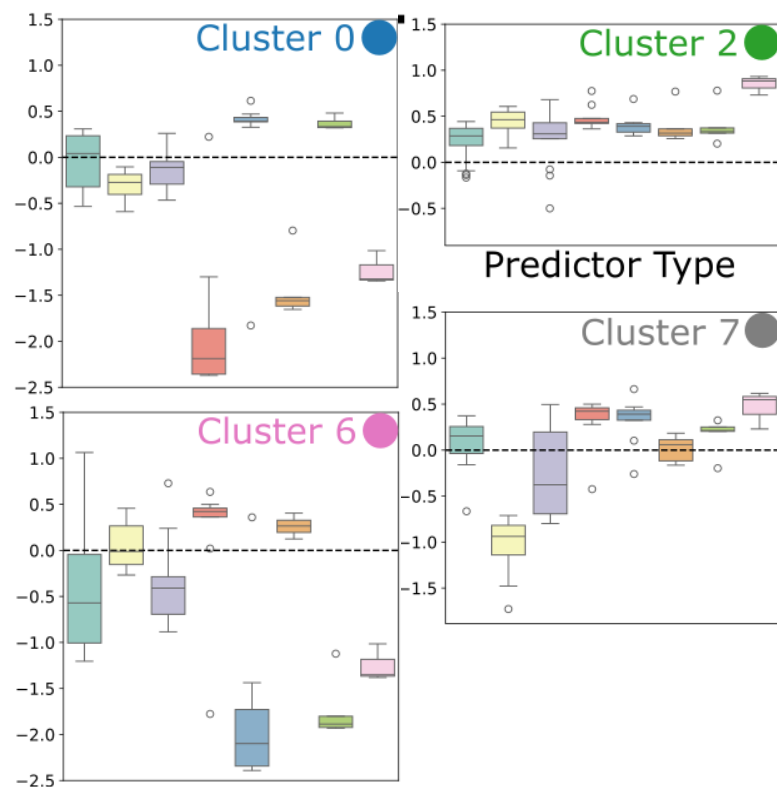
Can use Iowa Brain-Behavior Toolkit to perform lesion mapping analyses







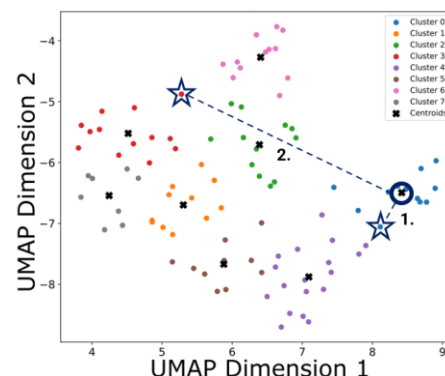
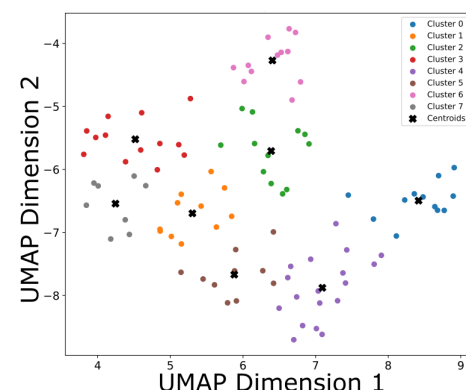
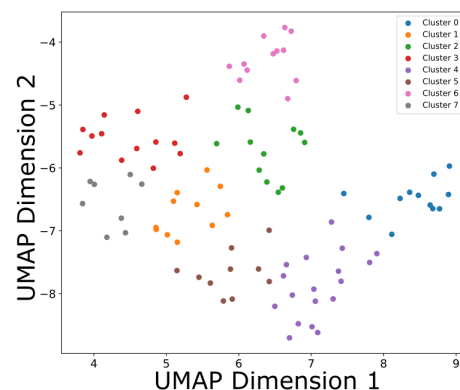
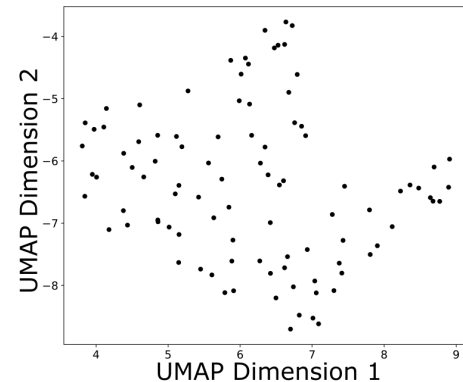




Processing Steps

1. Reduce test dimensionality with UMAP
2. Clustering on UMAP embedding
3. Calculate cluster centroids
4. Calculate patient distance to centroids
5. Perform lesion mapping with user's choice of toolkit

Patient ID	BNT (T_1)	...	T_N
001			
...			
P_N			



Patient ID	Dist. To Clu. k
001	
...	
P_N	

What we have done:

Outlined pipeline that could identify associations between a behavioral profile and lesion location

Using Behavioral Profiles for Prediction

Future directions



IOWA

Thank you!

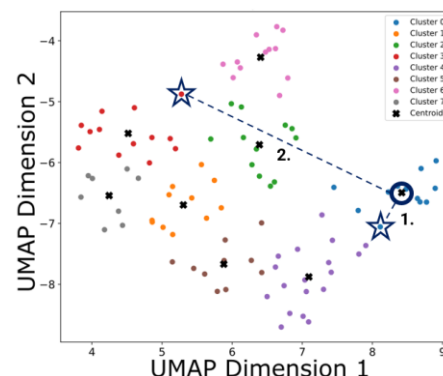
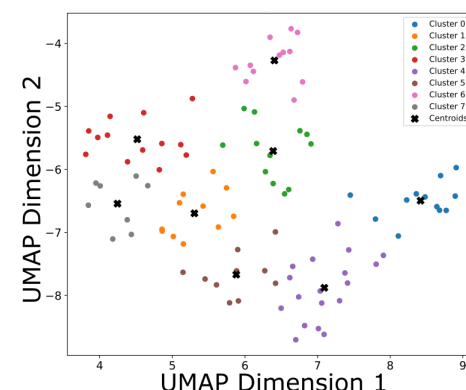
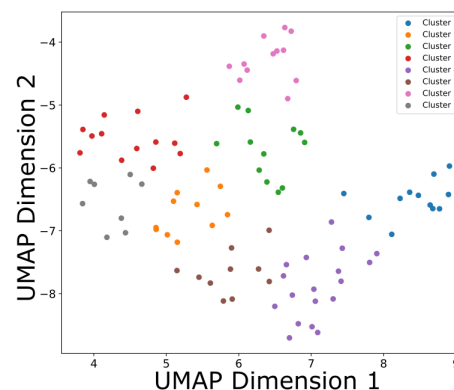
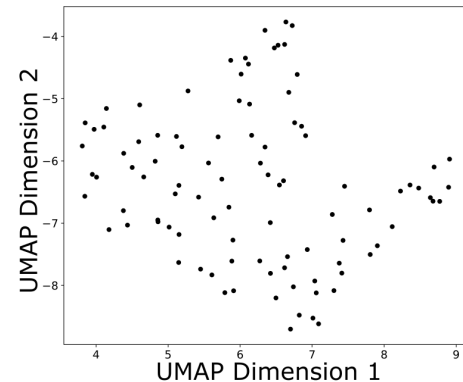
FUNDED BY:

Iowa Neuroscience
Specialty Program in
Research Education

Processing Steps

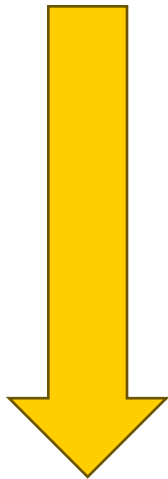
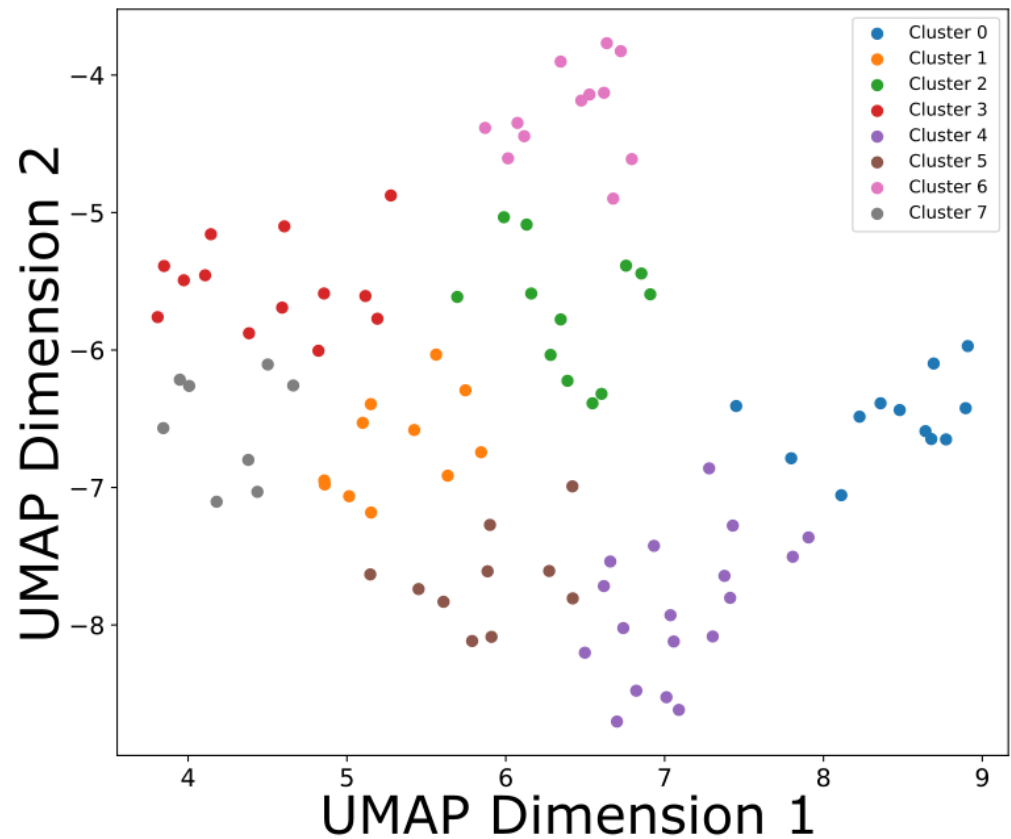
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Patient ID	BNT (T_1)	...	T_N
001			
...			
P_N			

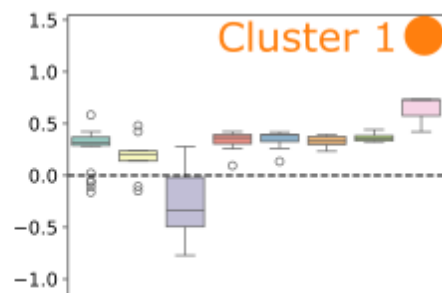
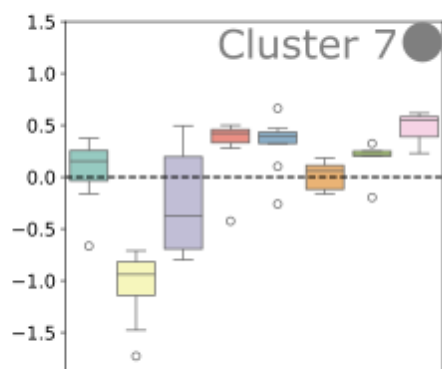
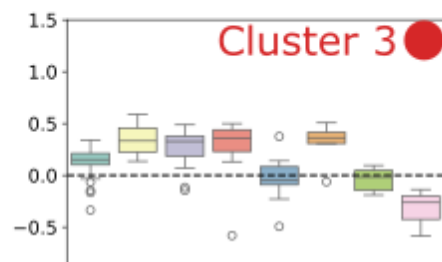
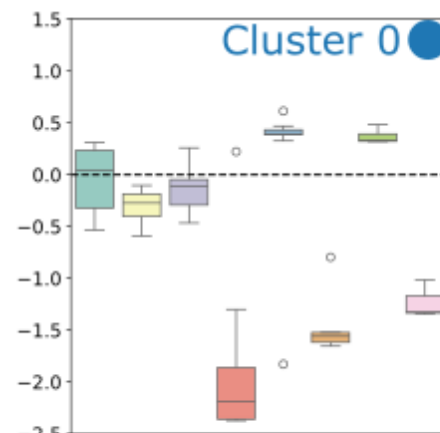
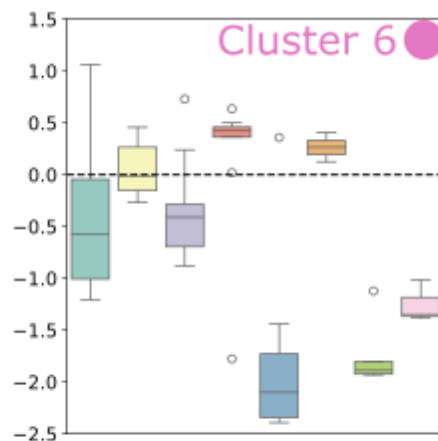
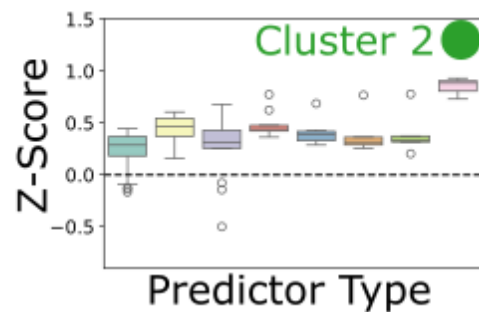


Patient ID	Dist. To Clu. k
001	
...	
P_N	

Clusters determined
quantitatively or
qualitatively



Average within-cluster
performance on each
test



KMeans Clustering of 2D UMAP (k=8)

